

SHRI VISHNU ENGINEERING COLLEGE FOR WOMEN :: BHIMAVARAM
(AUTONOMOUS)



ANNUAL REPORT 2023-24

SVECW IIC ID: IC201810226



A. About IIC Institute

The MHRD's National Innovation and Startup Policy is a guiding framework that needs to be put into place in institutions to help with actions related to innovation and entrepreneurship. According to this framework, the Institution Innovation Council of SVECW takes on the task of building an environment for innovation and entrepreneurship. This is made possible by VISHVA TBI, which was started at the college under NIDHI TBI, and the Entrepreneurship Development Cell.

VISION

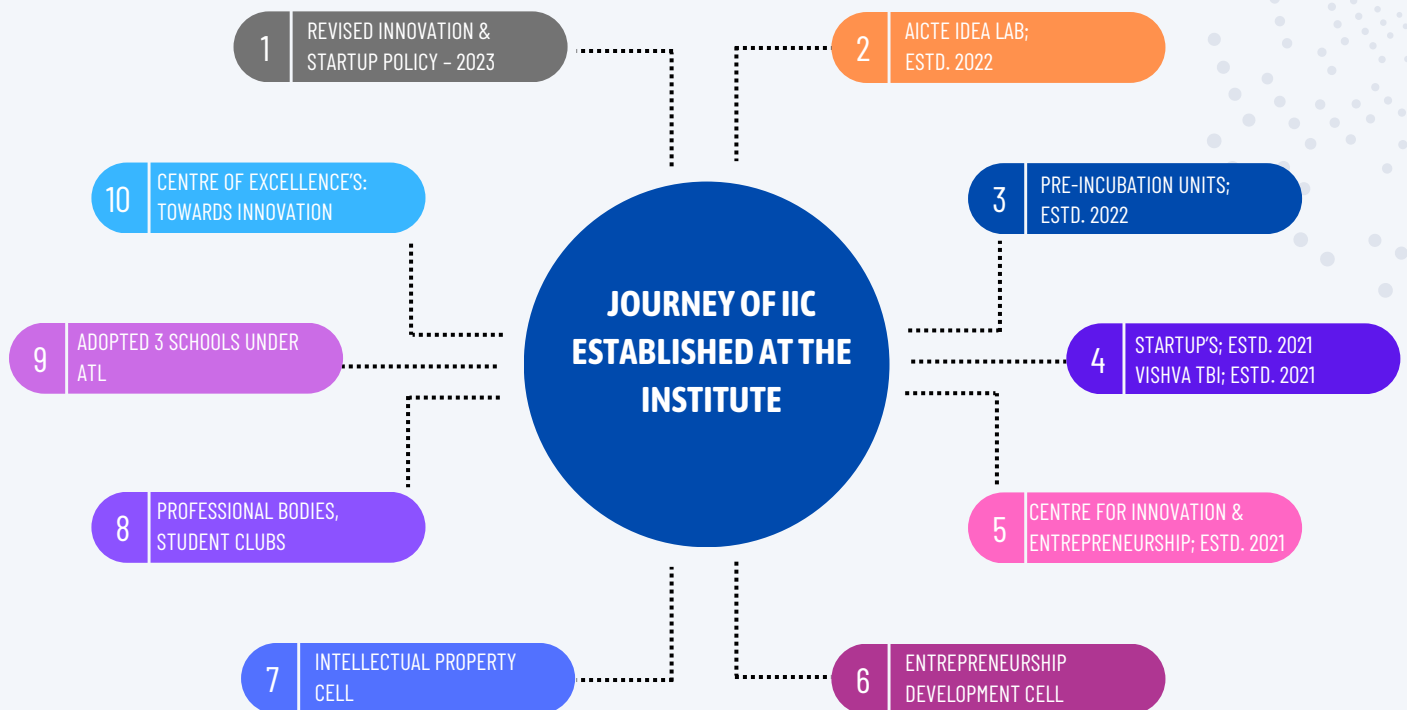
- To be recognized as a leading institution in the environment for promoting innovation and to spread the spirit of entrepreneurship in the campus.
- To make products and technologies that meet the needs of society by offering the innovative minds to be job providers

MISSION

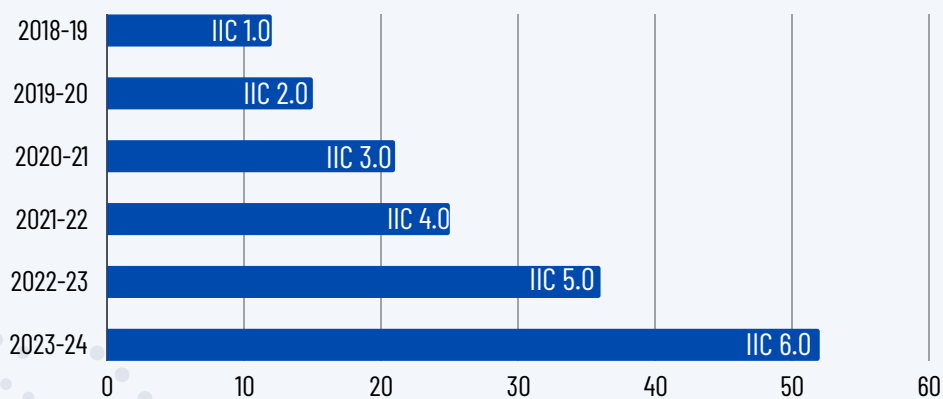
- Organise regular workshops, seminars, and interactive sessions with entrepreneurs, investors, and professionals to facilitate the dissemination and cultivation of creative ideas.
- Engage in professional networking activities with professionals and established national organisations focused on fostering entrepreneurship.
- Organize idea contests and Project Exhibitions to showcase the innovative ideas.
- Create an ecosystem that will foster entrepreneurial spirit among Faculty and Students. Offer mentorship to translate ideas into products and services that cater to the needs of the society.

Journey of IIC established at the Institute

The Institution's Innovation Council (IIC) is one of the Unique Initiatives of MoE's Innovation Cell, Govt. of India to promote Innovation, Start-up and Entrepreneurship activities. Our college registered for IIC in A.Y:2018-2019, to promote the vision of IIC in terms of Innovation, Start-up and Entrepreneurship among faculty members and students. College got a star rating of 3 out of 5 in IIC 4.0 and ranked in the band of 151-300 in the category of NIRF Innovation ranking 2023. Faculty and Students actively take part in various programs conducted by MIC, IIC and the college also conducts related activities on its own as self-driven activities.



A. Diversified representation in the IIC established at the institute from industry, Interdisciplinary & Departments/ Units etc.



IIC at SVECW represents the Innovation & Entrepreneurship training and promotion. Below is the representation of people from different centres and are part of IIC 5.0 for effective functioning.

- Dr. G. Subbaraju, IIC Convener
- Dr. K. Padma Vasavi, Coordinator IPR Cell & In-charge- NIDHI TBI Centre
- Dr. T Sudheer, IDEA Lab,
- Dr. V.V.R. Maheswara Rao, NIRF Coordinator
- Dr. G. Ratnakanth, Coordinator EDC
- Dr. G Durga Prasad, In-Charge Innovation Cell
- Mr. B Satya Krishna, In-Charge Startup Cell
- Mr. D. Ramesh Varma, Vishnu Foundation Technology Business Incubator
- Mr. Ch. Anudeep, Startup Activity Coordinator
- Mr. K. Pawan Kumar, Product Manager, Microsoft Corporation (I) Pvt. Ltd.
- Mrs. Shibani Pandit, Product Designer, VMware Software India Pvt. Ltd, Bangalore
- Mr. K. Prasanth, Co-Founder, StudyOwl, Bangalore
- Mr. Ravi Muttineni, Venture Capitalist, Bangalore
- Ms. Raga Mounika, Developer, Athena Health, Bangalore

- **Head of the Institute** – Mentoring and guiding to the IIC council members
- **President** – Conduction of Meetings, Discussion of events and activities to be conducted
- **Convenor** – Planning of activities according to IIC, uploading the reports received from coordinators, Updating the info to the faculty and students as soon as received from IIC
- **Coordinators and Members** – Conduction of activities and submitting the report

B. Brief mention of key functionaries at the IIC Institute



Sri K. V. Vishnu Raju
CHAIRMAN



Dr. G. Srinivasa Rao
HEAD OF THE INSTITUTE
(HOI)



Prof. P. Venkata Rama Raju
PRESIDENT



Dr. G. Subba Raju
Convener



Dr. K. Padma Vasavi
IPR Activity Coordinator



Dr. P. Kiran Sree
Social Media Coordinator



Dr. S.M. Padmaja
Innovation Activity Coordinator



Dr. S. Hanumantha Rao
Internship Activity Coordinator



Dr. V.V.R. Maheswara Rao
NIRF Coordinator



Dr. G. Durga Prasad
NISP Coordinator



Mr. Ch. Anudeep
Startup Activity Coordinator

C. Portfolio/ graphical/ Tabular representation of Resource strength (human capital and Physical capital) of the IIC institution

Total No. of IIC Members
Faculty:25, Students:24, External Members:3

Total No. of faculty Mentors
from Portal: 25

Incubation Units, If any: 1
(VISHVA TBI)



Total No. of IAs: 30
Faculty: (Reskilling: 3 || Advanced Level:11 || Basic Level:22)
Students: Basic level: 2

Pre-Incubation Units, If any :5
(Entrepreneurship development Cell, Innovation Cell, Startup Cell, NIDHI TBI, AICTE IDEA LAB)

IP Facilitation Unit, If any:
SVECW-IPR Cell

Innovation Ambassadors



Dr. K. Padma Vasavi
Prof. in ECE



Dr. P. Kiran Sree
Prof. in CSE



Dr. S. M. Padmaja
Prof. in EEE



Dr. VVR Maheswara Rao
Prof. in CSE



Dr. S. Hanumantha Rao
Prof. in ECE



Dr. G. Durga Prasad
Prof. in AI



Dr. S. D. K. Varma
Prof. in EEE



Dr. G. Ratnakanth
Prof. in IT



Dr. T. Sudheer Kumar
Prof. in ECE



Dr. M. V. V. Rama Rao
Asst. Prof. in CSE



Dr. G. Challa Ram
Asst. Prof. in ECE



Dr. D. Ramesh Varma
Asst. Prof. in ECE



Mr. Ch. Anudeep
Asst. Prof. in MBA



Mr. ASV Prasad
Asst. Prof. in ME



Mr. S. Ravi Chandra
Asst. Prof. in IT



Mrs. P. Archana
Asst. Prof. in AI



Mr. L. Ram Gopal
Asst. Prof. in CE



Mr. M. Rajesh
Asst. Prof. in ME



Mr. T. Sunil Kumar
Lab Technician in EEE



Mr. Sivaji
Lab Technician in CSE



Mr. P. Narasimharao
Lab Technician in ECE



Mr. N. Kalyan
Lab Technician in ME



Ms. Ch. Pallavi Sirisha
Student in ME



MS. K. Sri lakshmi
Student in ME

D. Highlight Facilities, Infrastructure of Pre-Incubation & Incubation kind and Student bodies / clubs engaged in promotion of Innovation and Entrepreneurship in the campus.

Technology Business Incubator (TBI)

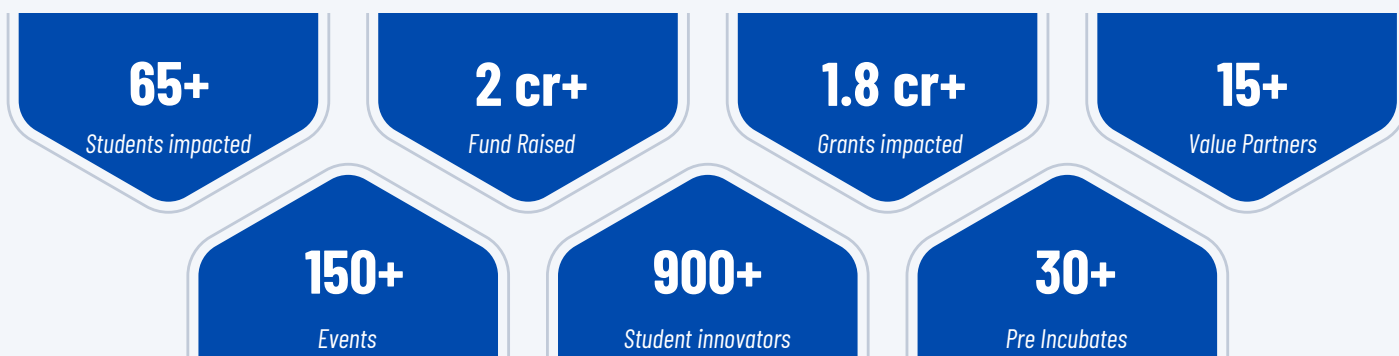


The Vishva Technology Business Incubator (TBI) at Shri Vishnu Engineering College for Women provides a dynamic platform for innovation and entrepreneurship. The Incubation Unit at Shri Vishnu Engineering College for Women actively nurtures student innovation by mentoring startups, organizing idea pitching sessions, and supporting proof of concepts (POCs) and product development. It also guides students toward incubation, trains them in business model canvases, and conducts entrepreneurial skills workshops. It fosters the development of cutting-edge technologies, supporting over 65 startups, with 12 of them securing external funding of Rs. 682 Lakhs. TBI has impacted more than 900 student innovators, offering pre-incubation to over 30 startups, including 42 with at-least one women founders. The incubator's initiatives span across various domains, encouraging innovation through events, mentorship, and access to resources. This year, TBI secured Rs. 5 Crores in funding under the Startup India Seed Fund Scheme.

Infrastructure @ TBI



Impact by Vishnu Foundation TBI



Innovation/prototypes developed





IDEA LAB

The purpose of IDEA Lab is to provide all facilities under one roof, for conversion of an idea into a prototype. With these facilities in the campus, more students and faculty will be encouraged to take up creative work and in the process, get training on creative thinking, problem solving, collaboration etc. which conventional labs are not able to. The focus will be on training students so that they become imaginative and creative and stay so at the workplaces they join. The whole idea is to transform engineering education with such a Lab in all colleges and for this they must proactively expose all students to the IDEA Lab, organize training sessions for interested students as well as support projects and by providing online learning materials.

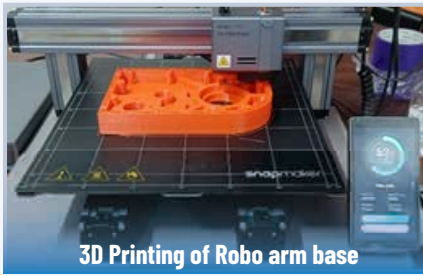
• Facilities @ IDEA Lab



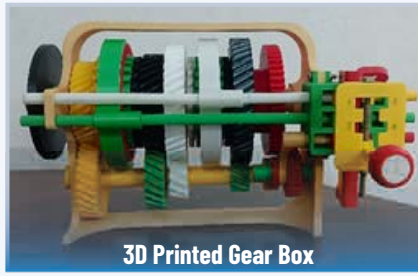
• Glimpses of the Events @ IDEA Lab



• Products developed @ IDEA Lab



3D Printing of Robo arm base



3D Printed Gear Box



3D Printed Crank Shaft



Replica of Chandrayan



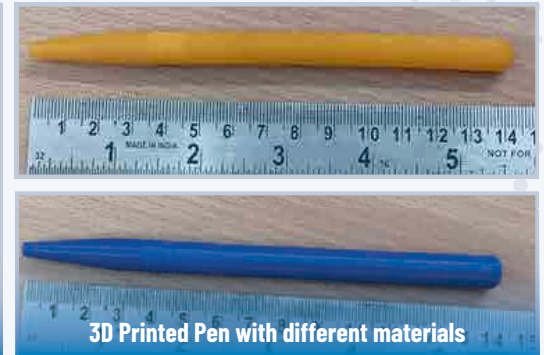
3D Printed Ramappa Temple



3D Printed 4Stroke 4Cylinder Engine



3D Printed Pick and Place ROBO



3D Printed Pen with different materials



3D Printed 4Stroke 4Cylinder Engine



3D Printed Phone Stand



3D Printed Ramappa Temple (Top view)

I&E COURSES

- Design Thinking
- Entrepreneurship and Small Business Management
- Organic farming
- Food Technology
- Intellectual Property Rights and Patents
- Design and Drawing of Reinforced Concrete Structures
- Innovations in Marketing

IPR CELL

- Innovation protection through Patent, Copyright and Trademarks
- Mentoring Support

EDC CELL

- Motivating Students Towards Entrepreneurship
- Mentoring Support
- Business Model Canvas

I&E ACTIVITIES

30

IIC Calendar Activity

06

MIC driven Activity

77

Self driven Activity

16

Celebration Activity

CENTERS OF EXCELLENCE :: TOWARDS INNOVATION

AR-VR Studio



On 7th August 2024, the AR & VR studio was inaugurated at Shri Vishnu Engineering College for Women, aiming to train students and faculty in Augmented Reality (AR) and Virtual Reality (VR) technologies. AR and VR present a wide range of opportunities for fostering a vibrant start-up ecosystem. These technologies are applicable across various industries, including gaming, education, healthcare, real estate, retail, and tourism, allowing start-ups to explore numerous niches and market segments.

Students can develop immersive user experiences, such as virtual tours, interactive training modules, and gamified learning environments. These innovations have the potential to disrupt traditional business models and open new revenue streams. For example, students can create and sell AR/VR applications or software solutions, including games, educational apps, training simulations, and productivity tools. Revenue can be generated through direct sales, subscriptions, or in-app purchases.

In addition, the subscription model for AR/VR applications or services offers ongoing access to premium content or features. This can include software-as-a-service (SaaS) platforms or subscription-based virtual environments, providing continuous opportunities for innovation and growth.

Vishnu Space Application Center (VSAC)

The Vishnu Space Application Center (VSAC) at Shri Vishnu Engineering College for Women (SVECW) stands as a beacon of innovation, offering women students unique opportunities to engage in cutting-edge space technology research. Through hands-on experience in satellite tracking, CubeSat design, and High Altitude Balloon (HAB) experiments, the center empowers students to develop technical skills essential for the rapidly growing space industry. By fostering a culture of innovation and collaboration, VSAC not only enhances the technical proficiency of students but also opens doors for entrepreneurship and startup ventures in the space tech domain, positioning SVECW as a hub for future women-led space startups.



Canoe Lab



The Concrete Canoe developed by students from the Department of Civil Engineering at Shri Vishnu Engineering College for Women (A) represents a remarkable blend of innovation and prototyping. Optimized in terms of materials, the canoe is crafted using lightweight, durable concrete, ensuring buoyancy while maintaining strength. The design has been carefully engineered for stability, making it capable of navigating various water conditions. Advanced construction methods have been employed to reduce material usage, minimize cost, and enhance durability. With a focus on both performance and affordability, this canoe prototype showcases the students' expertise in innovative material selection, structural design, and practical construction techniques. The project, developed under the Concrete Canoe Lab, is a testament to sustainable engineering practices in civil engineering education.

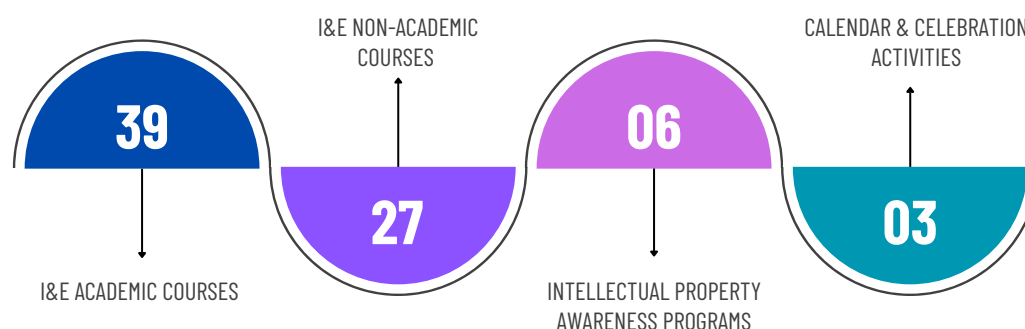
Professional Bodies

Student clubs



E. Highlight Achievements (Narrative/Graphical/tabular representation)

NUMBER AND DIFFERENT TYPES OF I&E AND IPR ACTIVITIES CONDUCTED



FEATURES	BASIC
No. of student's & faculty ideas generated	110
No. of student's & faculty Innovation/prototypes developed	99
No. of IPs generated, published and granted	PUBLISHED: 10 GRANTED: 14
No. of Student & Faculty Start-ups/Ventures established	03
Total Cost/Expenses spend towards innovation & Startups A) Total Program Implementation Cost: b) Total Infrastructure Development Cost: C) Total grant supported to innovation & startups Cost:	86.748 LAKHS 151.32 LAKHS 4.98 LAKHS
8) Sources of fund contribution to support innovation & startups: a) Total fund received from external agency & sources: B) Total fund spent from own resources:	214.75 LAKHS 19.32 LAKHS
No. of Technology Transfer and Commercialisation happened	12

F. Highlight few best IIC Faculty/Student members and their achievements / Rewarded for the innovations at different forum



Mr. G. Challa Ram | ECE

As an Innovation Ambassador at Shri Vishnu Engineering College for Women, I, Dr. G. Challa Ram, Assistant Professor in the Department of ECE, have greatly benefited from the Advanced IA training and reskilling provided by the Institute Innovation Council (IIC). These programs have not only enhanced my skills in fostering innovation but have also equipped me with the tools to guide students effectively in their entrepreneurial endeavors. Through the mentorship and support offered by Innovation Ambassadors, students at our institute are empowered to explore and develop their ideas, gain practical experience, and turn innovative concepts into impactful projects. This guidance helps them to navigate the challenges of entrepreneurship, preparing them for successful careers in an increasingly competitive world.

Dr. S. Dileep Kumar Varma | EEE



I Dr.S.Dileep kumar Varma Professor in EEE Dept. acting as an IIC Ambassador for Shri Vishnu Engineering College for women Bhimavaram. I feel honoured to be acted as an innovation ambassador for SVECW. I have organized several internal and external activities such as Project Expo, Ideation competitions, field visits, industry expert lectures and outreach programs at schools, to create awareness among students on Innovation, Entrepreneurship and start-ups. Moreover, I have delivered expert talks on innovation and design thinking at the institute and outside of the institute to empower the students and create interest among students on development of Innovative products for the benefit of community. As an innovation ambassador.

Mr. Ch. Anudeep | MBA



"For the past 6 years, I've had the privilege of serving as an Innovation Ambassador for the Institutes of Innovation Council. It has been incredibly fulfilling to guide and inspire students, helping them shift their perspectives on problem identification and problem-solving. Witnessing their growth and the innovative solutions they develop is a constant reminder of the power of creativity and critical thinking."

G. Highlight selected best Innovations & images with mention of inventor/innovation name

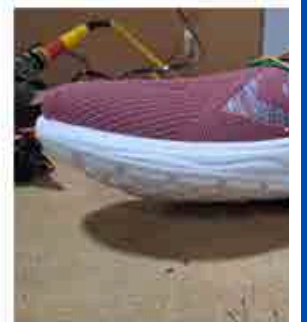
Name of the Project: AI Enhanced Road Image Annotation for Semantic Segmentation and Scenario Labelling

Ms. T. Navya Yasaswini, Ms. P. Jaswini, Ms. Md. Neha Taslim, and Ms. S. Gowthami, students of IV ECE, Shri Vishnu Engineering College for Women, achieved a significant milestone by being selected for the L&T Techgim 2024 Finals. Guided by Dr. M. Venkata Subbarao, they were one of top 32 teams chosen from over 4,400 teams across India, and the only team from Andhra Pradesh. Their pioneering project, 'AI Enhanced Road Image Annotation for Semantic Segmentation and Scenario Labelling,' was showcased at the prestigious L&T Techpark in Bangalore on 8th-9th May 2024. This remarkable accomplishment highlights their innovative approach and dedication to advancing AI technologies.



Name of the Project: Gyrosole

Gyrosole is developed by III ECE students of SVECW under the guidance G. Challa Ram. It utilizes advanced technologies, including pressure sensors to detect pressure at various points on the sole of the foot, a gyroscope sensor to track tilt and monitor gait and foot placement, and a GPS for tracking the user's location. This real-time monitoring system enables rapid response in emergencies by sending alerts to an administrator or tracker, ensuring the user's safety in dangerous situations.



Name of the Project: Campus Patrolling Electric Vehicle
(Retrofitting of Splender)

A Splender motorcycle has been successfully retrofitted from petrol to electric, replacing the internal combustion engine with a brushless DC motor and lithium-ion battery. The conversion includes a custom motor mount, battery enclosure, and electric throttle, resulting in a fully functional and eco-friendly electric vehicle. A splendor bike is retrofitted with 72v system of 3.2 kW battery capacity. The motor used is a BLDC motor of 2KW capacity. This bike is used for campus patrolling having 12v headlamp and required accessories. It is also having quick charge feature. It is having a range of 125 km



Name of the Project: TEAM SUSTAINABLE MUD BLOCKS (SMB)

TEAM SUSTAINABLE MUD BLOCKS (SMB) have been selected as one of the 75 winners of the BUILD program organized by IIT Hyderabad Technology Incubation Centre (iTIC) from the Civil Engineering department of SVECW(A). The Teams is currently under incubation and have received a seed money amount of one lakh rupees to produce an industry-ready prototype



H. Highlight selected start-ups established by students/faculties with mention of founder/cofounder name



Psyshell
Health Heroes LLP
Founder:
Namburi Rupa Sree



ESPORTSMATE
SOLUTIONS LLP
Founder:
Sai Anurag Ayyalasomayajula



Vishnuspire
Founder:
A Vishnu Vardhan

I. List if any break through Innovations / Technology Developed at the institute (2-3 technology with 2-3 lines about technology and innovation)



Name of the project: Smart Water Meter for Agriculture

K. Vidya Sowmika, Vadali Susmitha Sri Ramani and Ch.Gnanasri from III ECE have participated in the Build to Innovate contest conducted by IIT Madras Research Park. Total 18 teams from various institute have participated in this competition and three teams remains finalists. Our students stood at top three for the project of smart water meter for agriculture.

Name of the project: Intelligent Radiologist Assistant (Knee MRI)

The team of L. Rishitha, G. Sai Girija, G. Pravallika, and P. Joswitha from the Department of ECE, Shri Vishnu Engineering College for Women, has been selected for the semi-final round of L&T Techgim 2024. They showcased their project "Intelligent Radiologist Assistant (Knee MRI)," which utilizes cascaded deep learning architectures, specifically ResNet50 and YOLO, to improve the accuracy and efficiency of knee MRI analysis. This system aids radiologists by precisely detecting and diagnosing knee abnormalities, such as ACL tears or meniscal injuries, thus enhancing clinical outcomes and reducing diagnostic time.



Name of the project: Ecofriendly Concrete Canoe

Creating eco-friendly, cost-effective concrete canoe which can incorporate sustainable and recycled materials, reducing their environmental impact. They are non-combustible, providing an added safety benefit over wooden boats, which are susceptible to fire hazards.

J. Participation of IIC-institute in various programs of Central and Stage Govt. Highlighting specially for the schemes or programs

REGISTERED FOR ARIIA (AIR-C-17962)		APPROVAL OF INNOVATION AND ENTREPRENEURSHIP POLICY OF THE INSTITUTE	✓	IMPLEMENTATION ACTION PLAN DEVELOPED	✓	YUKTI INNOVATION CHALLENGE 2023	✓
NIRF INNOVATION RANK (2023)	151-300 BAND	POLICY NOTIFIED	✓	ALLOCATED 1% OF ANNUAL BUDGET	✓	IDE BOOT CAMP	✓
SMART INDIA HACKATHON	PARTICIPATING	DEDICATED POLICY WEB LINK	✓	CONDUCT OF AWARENESS PROGRAM ON POLICY OBJECTIVES, AND PROVISIONS	✓	ATL SCHOOLS	✓
KAVACH	✓	YUKTI NATIONAL INNOVATION CHALLENGE	✓	NEP KI SAMAJH - REG NEP 2020	✓	--	--

K. Detail of Social Media & Connections of IIC institute

- <https://www.linkedin.com/school/vishnusvecw/>
- <https://www.twitter.com/svecw2>
- https://www.instagram.com/vishnu_svecw/
- <https://www.facebook.com/svecwcollege>

L. Testimonials from IIC members and external about IIC institute and IIC of MoE's Innovation Cell



I Ravichandra Sriram, Assistant Professor, Dept of IT, IIC enables me to incorporate innovation and design thinking activities into their teaching, fostering interactive learning and equipping students with creative problem-solving skills.

I J Navya Sri, I am studying IIIrd year B.Tech IT, IIC provided the opportunity to participate in hackathons, design challenges, and innovation events, enhancing the creativity, problem-solving abilities, and practical experience in real-world scenarios.



I am T. Vasavi, studying IV EEE had the privilege of being actively involved in the Institution's Innovation Council (IIC) activities. This experience has provided me with immense learning opportunities, allowing me to engage in innovative practices that foster creativity, problem-solving, and leadership skills. Through various events and initiatives, I have enhanced my ability to think critically and contribute meaningfully to innovation-driven projects. My participation in IIC has not only deepened my understanding of entrepreneurship but also enabled me to collaborate with diverse teams, helping me grow both personally and professionally

I'm honoured to be a part of the Institutes Innovation Council. Engaging in innovation and entrepreneurship activities has been incredibly inspiring and motivating. It's an amazing opportunity that has significantly enhanced my skills and fuelled my passion for innovation



I'm grateful to be a part of IIC. Participating in innovation and entrepreneurship activities has been a rewarding experience. The opportunity for students to interact with renowned entrepreneurs is truly inspiring and encourages us to pursue our own entrepreneurial ambitions

I am Ms. Nanditha, a final-year ECE student at Shri Vishnu Engineering College for Women, and being a part of the Institute Innovation Council has been a transformative experience for me. The support and resources provided by IIC have been instrumental in my journey as an ECE student. I have had the opportunity to work on numerous projects, including reaching the finals of the L&T Techgrium contest and participating in multiple hackathons. I am grateful for the opportunities IIC has provided, which have greatly enhanced my skills and helped me achieve my goals.



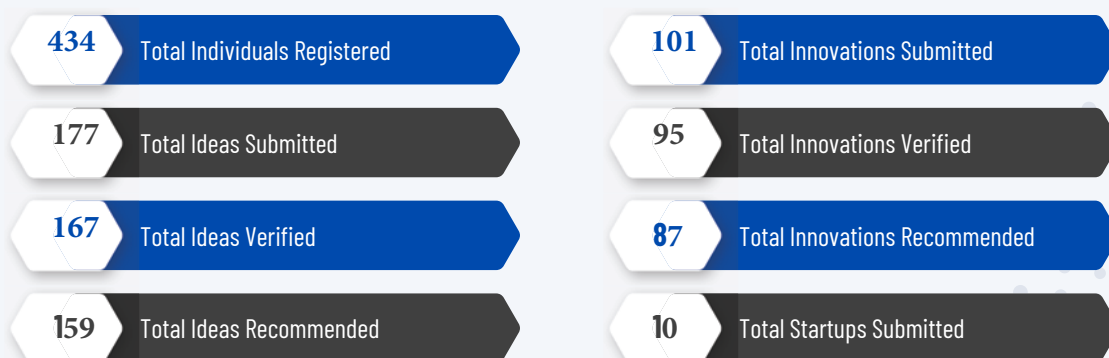
M. Images

Glimpses of the Events Organized by Various Departments under IIC:



Highlights of the Activities

YUKTI NATIONAL INNOVATION REPOSITORY



ATL School Mapping

Sl. No	School Name	E-Mail	Nominate IIC Coordinator
1	ZP HIGH SCHOOL, MOGALLU WEST GODAVARI	hmzphsmogallu@gmail.com	Dr. S.Dileep Kumar varma
2	ST ANNS ENGLISH MEDIUM SCHOOL WEST GODAVARI	venkatramanap69@gmail.com	Ch. Anudeep
3	VISHNU SCHOOL BHIMAVARAM	vishnupublicschool@yahoo.co.in	L. Ramgopal

Industry Partnership / Associations



N. Contact

Dr. G. Srinivasa Rao
Principal

Prof. P. Venkata Rama Raju
Vice-Principal
President - IIC

Dr. G. Subba Raju
Professor & HOD-MBA
Convener - IIC

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